

Restoring Connectivity Constraints in Joint UAV Trajectory and Handover Control

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Research draft, 12 September 2026

Abstract—Restoring destination arrival does not establish that a cellular UAV maintains its required connection. We audit an earlier reward repair and implement a separate study with strict sampled received signal strength and buffer constraints, together with the original thesis’s delay, interference and handover objectives. A received Barcelona radio map supplies the channel data. All new controllers share explicit surrogate dynamics and a prospective safety filter; learned controllers add bounded motion corrections to a braking reference and select among available station/resource actions. Two reward arms use five training seeds and fresh standard and longer route sets. Joint mission success is 96.6% and 89.7% with the full radio cost, versus 96.9% and 86.3% without that cost. On paired successful standard flights, accumulated radio cost changes by -24.5%, while the mean delay proxy changes by 186.7%. Four deterministic references expose the contributions and limits of network control and local route planning. Every reported successful flight satisfies the sampled RSS and buffer constraints. This repairs the objective and endpoint within the declared surrogate; it does not establish continuous physical connectivity, improvement in every communication metric, or reproduction of the original simulator.

Index Terms—Cellular UAV, connectivity constraints, handover, trajectory control, reward design, reinforcement learning, reproducibility.

I. INTRODUCTION

A master’s thesis considers joint trajectory and handover control using a Barcelona ray tracing environment [1, pp. 6–7, Sec. 3.1]. Its objective combines delay, interference and handovers [1, p. 10, Eq. (11)], while its constraints require minimum serving RSS and bounded queue occupancy [1, p. 11, Eq. (12), C2 and C4]. The source nevertheless reports wandering and prioritization of connectivity over destination completion [1, p. 19, Sec. 7.2; p. 21, Sec. 8].

An earlier controlled reimplementation replaced a binary approach reward and restored arrival on the received map. That finding had a narrower scope than the full joint task: mission success permitted brief RSS outages, communication feasibility was partly a subsequent label, and the replacement reward omitted direct interference cost. A follow up audit motivated the present study.

We correct the sampled joint endpoint and objective through a controlled comparison with retained development failures. The original simulator remains unavailable. Citations give printed thesis pages; add two for the PDF viewer page. Accompanying Markdown notes contain complete implementation comparisons.

II. SOURCE REQUIREMENTS AND AUDIT

A. Which Communication Requirements Apply?

We use the thesis requirements, not a newly imposed application service level. The RSS minimum is -96 dBm and the buffer is listed as 160 KB [1, p. 15, Table 3]; we retain the explicit 1,280,000 bit interpretation. The written C2 inequality allows equality and applies at every modeled time [1, p. 11, Eq. (12)]. Delay remains an optimized quantity, defined in the source as queue divided by service rate [1, p. 9, Eq. (4)]. Neither a two second packet deadline nor an empty queue arrival condition is introduced. A successful arrival can therefore retain queued data; it must not be described as completed data delivery.

The distinction between formulation and enforcement matters. The source reward penalizes RSS violations [1, p. 12, Eq. (14)], which does not establish that its unavailable implementation guaranteed C2. Its direct data dump penalty was removed during development [1, pp. 20–21, Sec. 7.3]. We compare verified written definitions with our executable rules, not inferred behavior of missing original code.

B. Why the Earlier Arrival Result Was Insufficient

The original map audit found at least 38 operator 1 stations strictly above -96 dBm at every grid location. The strongest station ranges from -51 to -25 dBm. Consequently, an RSS coverage hole does not force a detour if any suitable station could be selected. Actual association constraints and interference still matter.

An exact replay of 4,400 earlier flights added service diagnostics without changing their saved outcomes. In the earlier reward replacement’s 1,000 standard evaluations, arrival was 100%, but only 92.2% had no sampled RSS outage. Modeled capacity was below the 200 kbit/s offered rate for a mean 26.6% of each flight, and 20.2% arrived with a nonempty queue. A transient capacity deficit is not itself disconnection because buffering can absorb it. These measures expose what the old endpoint did and did not establish; they are not a new confirmatory policy comparison.

III. REVISED SURROGATE AND CONTROLLER

A. Shared Physical Assumptions

We preserve the received native resolution RSS map, selecting the first operator’s 87 stations. Matching source metadata includes 2.1 GHz and 100 m altitude [1, p. 7, Table 2; p. 15,

Table 3]; this does not prove an identical original dataset version. The raw private HDF5 is not modified or published. The provisional -128 sentinel interpretation remains unverified.

The shared model retains a one second decision step, acceleration norm limit 5 m/s^2 , speed limit 25 m/s , 200 s deadline, and a 100 kJ budget under the engineering energy proxy $100 + 0.4v^2$ watts. These are the earlier surrogate's settings. They differ from the source's 0.1 s table step and 100 m/s speed setting [1, p. 15, Table 3]; the source also gives a conflicting 1 ms interval in prose [1, p. 14, Sec. 6.1].

The radio proxy retains 12 resource groups, half occupied by a deterministic station pattern. Interference sums linear received power from cochannel background stations. Capacity is $1.44 \cdot 10^6 \log_2(1 + \text{SINR})$ bit/s, with noise -103.41 dBm and offered traffic $200,000 \text{ bit/s}$. The source's printed rate and SNR expressions differ [1, p. 10, Eqs. (7)–(8)], and its background occupancy is described as random [1, p. 14, Sec. 6.1]. We retain these declared surrogate assumptions to separate the current repair from a new propagation or traffic model. The interference is a downlink proxy, not a calibrated uplink quantity matching the source's terminology [1, p. 10, Eq. (9)].

B. Joint Success and Failure

Let d_t denote destination distance and q_t the queue after the sampled arrival/service update. Joint success requires arrival by the deadline with $d_T \leq 10 \text{ m}$ and $\|v_T\| \leq 2 \text{ m/s}$, together with

$$\text{RSS}_{k_t}(p_t) \geq -96 \text{ dBm}, \quad 0 \leq q_t \leq 1,280,000 \text{ bits}, \quad \forall t. \quad (1)$$

The 10 m tolerance follows the source table [1, p. 15, Table 3]; the numeric stopping tolerance is our explicit interpretation of stopping at the destination [1, p. 9, Sec. 3.2]. Map and energy conditions also apply. A single sampled RSS violation or any buffer overflow ends a failed mission. Failure takes precedence over simultaneous arrival. Initial RSS is checked; queue clipping for storage cannot erase an overflow flag.

This enforces the constraints at modeled samples. It is not a guarantee between the one second samples or within the queue update. No minimum SINR, packet latency threshold, or final queue draining requirement is claimed.

C. Actions, Residual Motion and Safety Filter

The source includes BS and RBG actions [1, p. 12, Sec. 5.2]. We expose 61 categorical options: stay, or five candidate station slots by 12 RBGs. Candidates are serving plus the four strongest RSS stations. Masks reject occupied groups and duplicated serving station slots. A new station must have current RSS more than 3 dB above the serving RSS and meet the minimum, following the source's A3 direction [1, p. 9, Eq. (3); p. 15, Table 3]. Network decisions occur each modeled second, replacing the earlier surrogate's five second interval. Resource changes within the same station are allowed and counted separately from handovers. Handovers generate $4,800$ control bits [1, p. 15, Table 3]; switching interruption is not modeled.

The observation has 156 features covering geometry, velocity, time, queue, energy, serving SINR, candidate RSS and positions, option capacities and masks. Two Gaussian outputs define bounded residual acceleration commands. We project a destination directed braking command onto the unit disk, add half of the tanh transformed residual, and apply the shared acceleration projection. Both learned arms use this prior; they do not learn basic flight entirely from scratch. The source's fully discrete action and smaller state differ [1, p. 12, Sec. 5.2].

A shared safety filter predicts the proposed next position, RSS and queue. It first tries another admissible station/resource pair for the same motion. If none is feasible, it invokes the local motion/network search. If no candidate repairs the step, the flight fails normally. The filter has prospective access to the known map and is an explicit control component, not learned competence. Interventions are recorded. It cannot establish global reachability.

D. Restoring the Communication Objective

Using source equal weights and scales [1, p. 15, Table 3], define

$$C_t = 0.35 \frac{10D_t}{1 + 10D_t} + 0.30 \frac{10^5 I_t}{1 + 10^5 I_t} + 0.35 \frac{100H_t}{1 + 100H_t}. \quad (2)$$

Here D_t is the backlog/service delay proxy in seconds, I_t interference in watts, and H_t a handover indicator. These are one minus the source reciprocal utilities [1, p. 12, Eq. (13)], removing the positive reward for merely remaining active while retaining all communication components. The earlier replacement's flat delay penalty above one second is removed. The reciprocal cost still saturates smoothly, which can favor substantial delay in exchange for fewer handovers; it is not a latency guarantee.

Both learned arms use

$$\Phi_t = -\frac{d_t + \|v_t\|^2/(2a_{\max})}{100 \text{ m}}, \quad (3)$$

with terminal potential zero. The base reward is $0.99\Phi_{t+1} - \Phi_t - 0.05$, plus 20 for joint success and minus 20 for terminal failure. The arrival arm uses this base alone; the full arm subtracts C_t . Both have identical hard failure rules and control interfaces. This is not a comparison to the original trained thesis policy or a claim that a finite scalar reward proves optimal lexicographic control.

IV. EXPERIMENTAL PROTOCOL

A. Learning, Splits and Freeze

PPO [2] uses separate two layer 64 unit tanh actor and critic networks, 64 lanes, 128 step rollouts, four epochs, minibatches of 512, learning rate $3 \cdot 10^{-4}$, discount 0.99, GAE 0.95, clip 0.2, entropy weight 0.005, value weight 0.5 and gradient norm limit 0.5. Running return variance normalizes rewards, which are clipped to $[-10, 10]$. Likelihoods refer to proposed residual and network actions; projection and filtering are environment transformations.

TABLE I
JOINT SUCCESS AND SUCCESSFUL STANDARD MISSION COST. LEARNED
ROWS AGGREGATE FIVE SEEDS ON 200 ROUTES; REFERENCES USE 200
ROUTES.

Controller	Standard (%)	Longer (%)	$\sum C_t$
PPO arrival	96.9	86.3	6.57
PPO full	96.6	89.7	4.97
Goal RSS	97.0	87.0	6.22
Goal radio	98.0	92.5	3.23
Joint one step	96.5	90.0	2.62
Joint three steps	95.0	90.5	1.92

Training has 4,096 routes of lengths 200–1800 m. Validation has 64 standard and 32 longer routes. Fresh tests contain 200 standard routes of 200–1000 m and 200 longer routes of 1000–1800 m. Exact route pairs do not overlap with the earlier study or other v2 splits, but all use the same geographic map. Both length ranges occur during training, so the longer set is not an extrapolation test. The source does not document equivalent three way separation [1, pp. 15–16, Secs. 6.2–6.3].

Six development policies and unsuccessful local planning variants are retained. Final source, scenario and protocol hashes were frozen before test evaluation. Each arm uses seeds 2101–2105 and 524,288 interactions, totaling 5,242,880 final interactions across ten policies. Evaluation uses the final checkpoint and deterministic actions; no test result selects a checkpoint or controller revision.

B. References and Analysis

Four deterministic references share the same filter and rules: nominal straight flight with strongest RSS association (Goal RSS); the same nominal motion with prospective station/resource cost minimization (Goal radio); a one step joint search; and a three step joint search. Joint searches use nine initial motion primitives: direction offsets and speed reductions of the braking reference. They prioritize sampled feasibility and progress, then communication cost. The three step search follows each initial choice with two predicted reference steps. Both are local methods and may fail. The filter can alter nominally straight motion, which is reflected in actual paths and intervention counts.

Primary reporting includes every mission and its failure reason. Secondary communication comparisons use the intersection of successful paired routes; intersection sizes and accumulated and average costs are retained. Paired differences use 5,000 crossed bootstrap draws over seed identities and shared route identities; deterministic comparisons resample routes only. Five training seeds do not represent independent geographic or traffic realizations. The prespecified strong joint improvement claim requires a positive lower interval for completion improvement and a negative upper interval for accumulated communication cost change. Individual components are reported regardless.

V. RESULTS AND INTERPRETATION

A. Strict Joint Completion

Table I and Fig. 1 report the frozen results. The full learned arm achieves 96.6% standard and 89.7% longer joint success, compared with 96.9% and 86.3% for the arrival arm. The paired standard completion difference is -0.3 percentage points, with 95% interval [-2.6, 2.0]. The corresponding longer difference is 3.4 points [-1.0, 8.1].

Both completion intervals include zero. The prespecified strong joint improvement claim is therefore not met on either split, despite lower weighted communication cost. Goal radio has the highest observed completion among the tested controllers: 98.0% standard and 92.5% longer. These results do not establish a PPO advantage over the simpler reference.

Every reported success has zero observed RSS violations and zero overflow, as required by the new endpoint. This does not mean every evaluated flight succeeds. The full arm has 34 failed standard evaluations and 103 failed longer evaluations out of 1,000 in each set. Failure reasons and all per seed outcomes remain in the analysis artifacts.

B. Communication Tradeoffs

There are 945 paired successful standard evaluations and 802 longer evaluations for the learned comparison. On those pairs, accumulated communication cost changes by -24.5% and -14.7%, respectively. Table II shows the individual components and uncertainty. An improvement in the weighted objective is not equivalent to improvement in every metric. In particular, the standard mean delay proxy changes by 186.7%, while handovers change by -95.5%.

Goal radio improves longer completion over Goal RSS by 5.5 points [1.5, 9.5] and reduces accumulated cost by 41.0% on 171 paired successful routes. Three step planning reduces cost by a further 42.9% relative to Goal radio on 173 longer pairs, but its completion difference of -2.0 points [-6.5, 2.5] does not establish improvement.

On the matched standard routes, the full arm’s delay proxy is 4.21 s versus 1.47 s for the arrival arm, while handovers fall from 8.55 to 0.38 per mission. On matched longer routes, delay is 6.62 s versus 1.83 s. The unweighted interference difference is inconclusive on both splits. These component results limit the interpretation of the favorable aggregate cost.

C. What Is Fixed, and What Remains Open?

The corrected implementation no longer counts a sampled RSS or buffer violation as a successful joint flight. Delay, interference and handovers are active, and admissible RBG choices are explicit. This repairs the earlier formulation and evaluation omission without making curvature an objective or guaranteeing success on every feasible mission.

The source’s reciprocal tradeoff can tolerate substantial delay and the stated contract permits a remaining queue at arrival. An application deadline would require a separately declared requirement, not a retrospective threshold.

Dynamic load, prediction errors, switching interruptions, finer time resolution and independent maps remain untested.

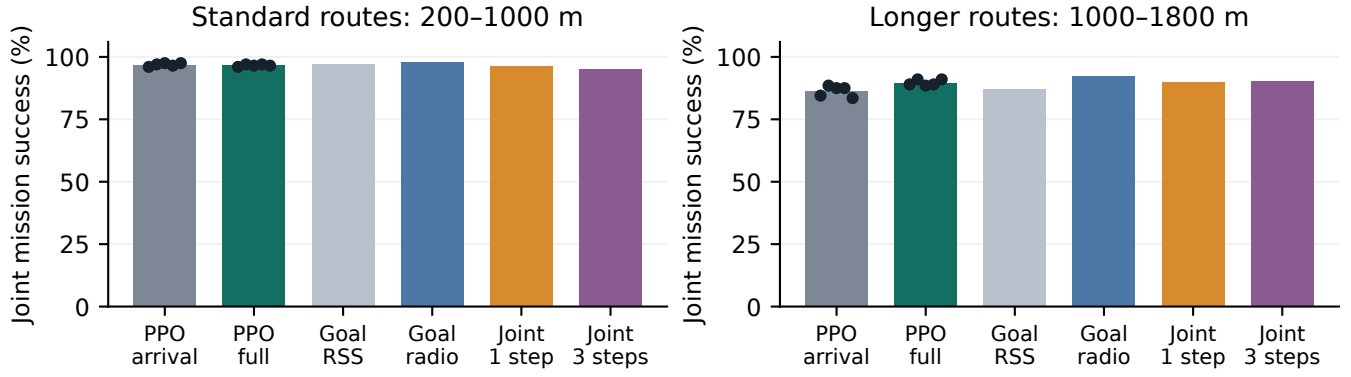


Fig. 1. Strict sampled joint success on fresh routes. Dots show individual training seeds for learned controllers. Deterministic controllers have one evaluation per route, not five independent replications.

TABLE II
FULL MINUS ARRIVAL ON PAIRED SUCCESSFUL MISSIONS. DIFFERENCES AND 95% CROSSED BOOTSTRAP INTERVALS. NEGATIVE COST, TIME, DELAY, INTERFERENCE AND HANDOVER CHANGES FAVOR FULL.

Metric	Standard difference [95% interval]	Longer difference [95% interval]
Accumulated radio cost	-1.610 [-2.016, -1.230]	-2.393 [-3.317, -1.488]
Mean radio cost	-0.060 [-0.075, -0.046]	-0.039 [-0.054, -0.024]
Arrival time (s)	-0.195 [-0.267, -0.124]	-0.353 [-0.552, -0.168]
Delay proxy (s)	2.738 [2.030, 3.474]	4.789 [4.180, 5.397]
Interference (μ W)	0.007 [-0.005, 0.018]	-0.002 [-0.012, 0.008]
Handovers	-8.165 [-9.226, -7.155]	-19.562 [-21.719, -17.263]
Mean SINR (dB)	-1.095 [-1.374, -0.801]	-1.581 [-1.780, -1.393]

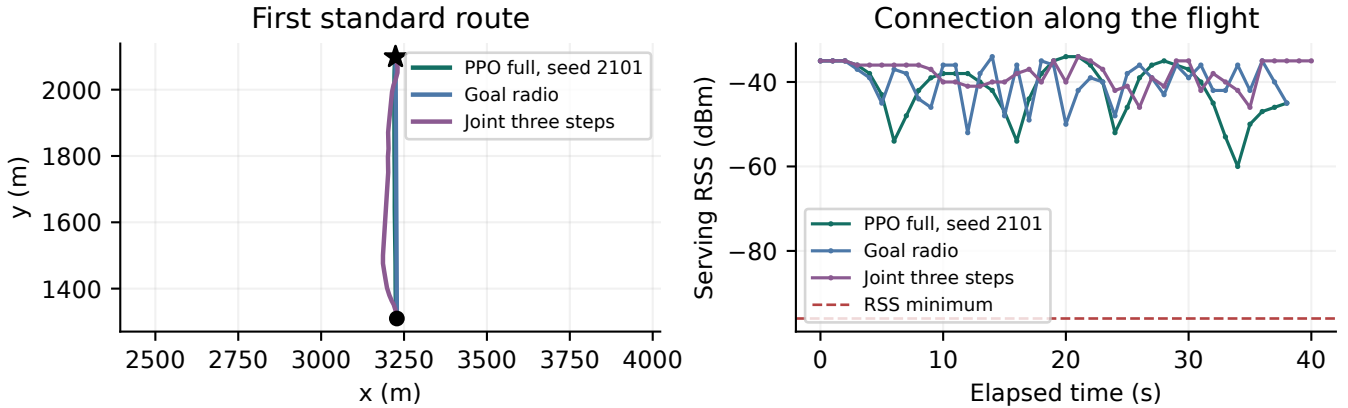


Fig. 2. The first declared standard test route and first full reward training seed, shown with two model based references. The displayed RSS includes the initial and terminal sample. This is a fixed example, not a selected typical route or proof of connectivity between samples.

The motion prior and expanded training range prevent attribution of v2 versus v1 differences solely to reward. Only the two v2 learned arms isolate the radio cost within their shared architecture.

VI. REPRODUCIBILITY AND CONCLUSION

The package retains hashes, routes, ten final and six development checkpoints, raw outcomes, paired analysis, and Markdown notes with thesis page comparisons. All 51 tests pass and all 5,600 final evaluations replay exactly. The dataset and both experiment freezes remain unchanged.

The repair makes communication requirements explicit alongside arrival. Lower weighted cost coexists with higher

delay and uncertain completion improvement. Remaining failures and surrogate assumptions preclude a claim of complete physical connectivity or general learned controller superiority.

REFERENCES

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